

Aayush

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LeetCode: 400+

Profile Summary

B.Tech CSE student (CGPA 9.43/10, graduating 2027) with strong Computer Science fundamentals and a solid hold on data structures, algorithms, and coding (400+ LeetCode problems solved). Hands-on experience designing, developing, and debugging production-grade ML/AI systems in Python – from-scratch model implementations, fine-tuned transformers, and optimized retrieval pipelines. Strong mathematical grounding in probability, statistics, and linear algebra. Seeking a Software Development Engineer Internship to build and ship scalable, reliable software in a collaborative, fast-paced engineering team.

Education

Bachelor of Technology in Computer Science and Engineering 2023 – 2027
I.K. Gujral Punjab Technical University CGPA: **9.43** *Relevant Coursework: Data Structures & Algorithms, Probability & Statistics, Linear Algebra, Optimization, DBMS*

Projects

ML Algorithms from Scratch using NumPy *

- Designed and implemented Linear Regression, Logistic Regression, and a feedforward Neural Network from first principles in Python/NumPy; manually derived gradient descent and backpropagation, including MSE/Cross-Entropy losses and L2 regularization
- Benchmarked against Scikit-learn on Iris and Boston Housing datasets, achieving within 2% accuracy; validated generalization via k-fold cross-validation and bias-variance tradeoff analysis
- Tech Stack:** Python, NumPy, Matplotlib, Gradient Descent, Backpropagation, Cross-Validation

Enterprise RAG AI Assistant *

- Architected and deployed a production-grade RAG web application (Streamlit) over 1,000+ PDF pages; diagnosed and optimized a FAISS vector pipeline across 50,000+ embeddings, cutting retrieval latency from 4.8s to under 1.5s
- Implemented conversational memory, improving multi-turn query accuracy by 35% over baseline LLM responses; supported 10+ file formats with citation-aware responses
- Tech Stack:** Python, SQL, LangChain, FAISS, ChromaDB, Gemini/OpenAI API, Streamlit, Vector Databases

Sentiment Analysis with Fine-Tuned DistilBERT *

- Fine-tuned DistilBERT for binary sentiment classification on 50K+ reviews; achieved 91% F1-score (5-fold CV), a 15% improvement over TF-IDF + Logistic Regression baseline
- Built and debugged a full PyTorch training pipeline with tokenization and confusion-matrix evaluation; published model on Hugging Face Hub
- Tech Stack:** Python, PyTorch, TensorFlow, Hugging Face Transformers, Scikit-learn, Pandas, NLP

Skills

- Programming & Tools:** Python, Java, C++, SQL, MySQL, Git, Linux/Bash, Jupyter, Kaggle
- ML & Deep Learning:** TensorFlow, PyTorch, Supervised/Unsupervised Learning, Gradient Descent, Backpropagation, CNNs, Transformers, Random Forest, XGBoost, Feature Engineering, Cross-Validation
- GenAI & NLP:** Hugging Face Transformers, RAG, FAISS, Embeddings, LangChain, Prompt Engineering
- Mathematics for AI:** Probability, Statistics, Hypothesis Testing, Linear Algebra, Optimization, Calculus
- Data Science:** NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, EDA, Data Preprocessing

Achievements & Leadership

- Solved **400+ DSA problems** on LeetCode with an active streak; strong focus on graphs, dynamic programming, and binary search
- Kaggle Competition — Titanic: Machine Learning from Disaster** — Achieved top 8% ranking (global leaderboard) with an ensemble of Random Forest + XGBoost and advanced feature engineering
- District Topper** in a National-Level Examination; **Ranked 1st** in Robotics and Artificial Intelligence branch
- Campus Executive, **E-Cell IIT Bombay** — led cross-functional teams for technical events and workshops
- Member, **Training Advisory Council (TAC)** — collaborated with faculty on curriculum development initiatives

Certifications

- Building AI-Powered Search with MongoDB Vector Search** – MongoDB *
- LLMs on Google Cloud: Retrieval-Augmented Generation on Vertex AI** – Infosys *
- Introduction to Natural Language Processing** – Infosys *
- AWS Academy Graduate - Generative AI Foundations - Training Badge** – AWS *